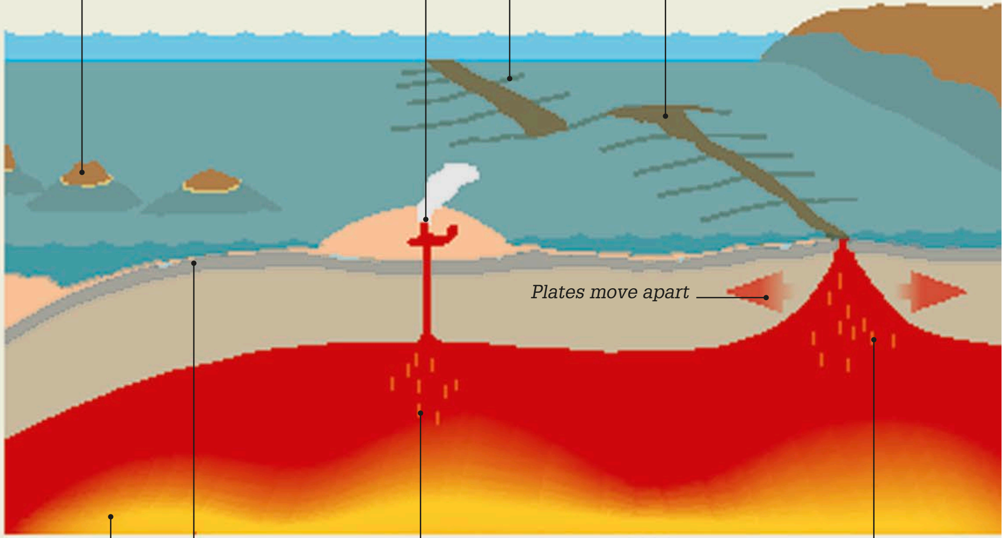
Plate tectonics

Earth's crust is made up of huge slabs of rock called tectonic plates that move constantly across the surface of the planet. When two plates move together, it causes Earth's crust to buckle, forming huge mountain ranges. Where plates move apart, it creates a rift (crack) in the crust. Here, molten rock, called magma, erupts from the mantle to form new ocean floors and ridges.

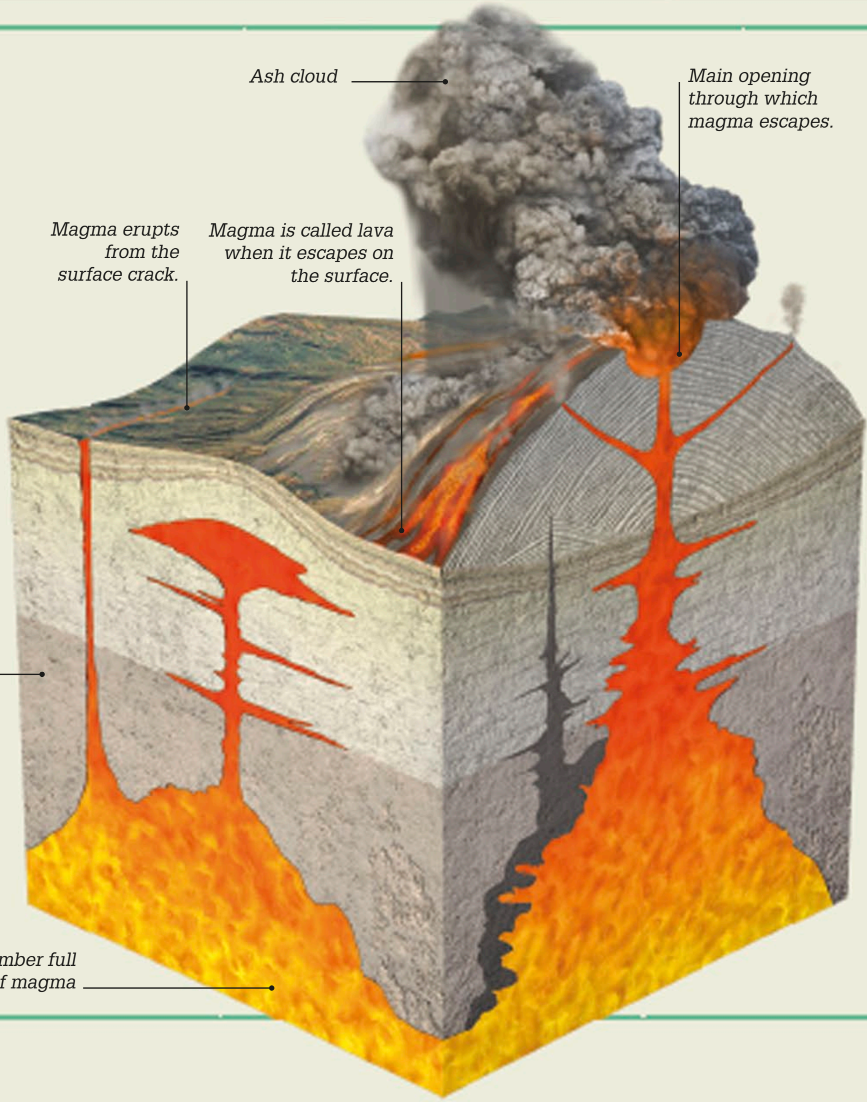


Violent Earth

Most earthquakes and volcanoes occur at plate boundaries where the tectonic plates collide, rub together, or move apart. When plates grind against each other, earthquakes occur as the rocks catch and then jerk free. When the plates move apart, magma from the mantle rises through weak points in Earth's crust and erupts at the surface as a volcano.

Volcanoes

Deep inside Earth, rocks melt into a hot, thick liquid called magma (molten rock). Now and again, this magma surges up to Earth's surface and pours out in a volcanic eruption. In some places, the molten rock (known as lava once it reaches the surface) oozes out slowly. In others, the eruption is a violent explosion of lava, red-hot lumps of rock, and clouds of scorching ash and steam. Some volcanoes are very active, while others erupt only rarely.



How volcanoes work

A volcano is a vent where magma from the Earth's hot interior emerges onto the surface. Volcanoes erupt violently when the build-up of magma and gases in the chamber below the vent creates enough pressure to blast through to the surface. The lava cools and sets, shaping the volcano.